Optimal Power Transmission, Interface Management and Congestion Analysis (OPTIMA)

06th Apr 2026

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Overview of OPTIMA Application

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1. Overview of OPTIMA Application

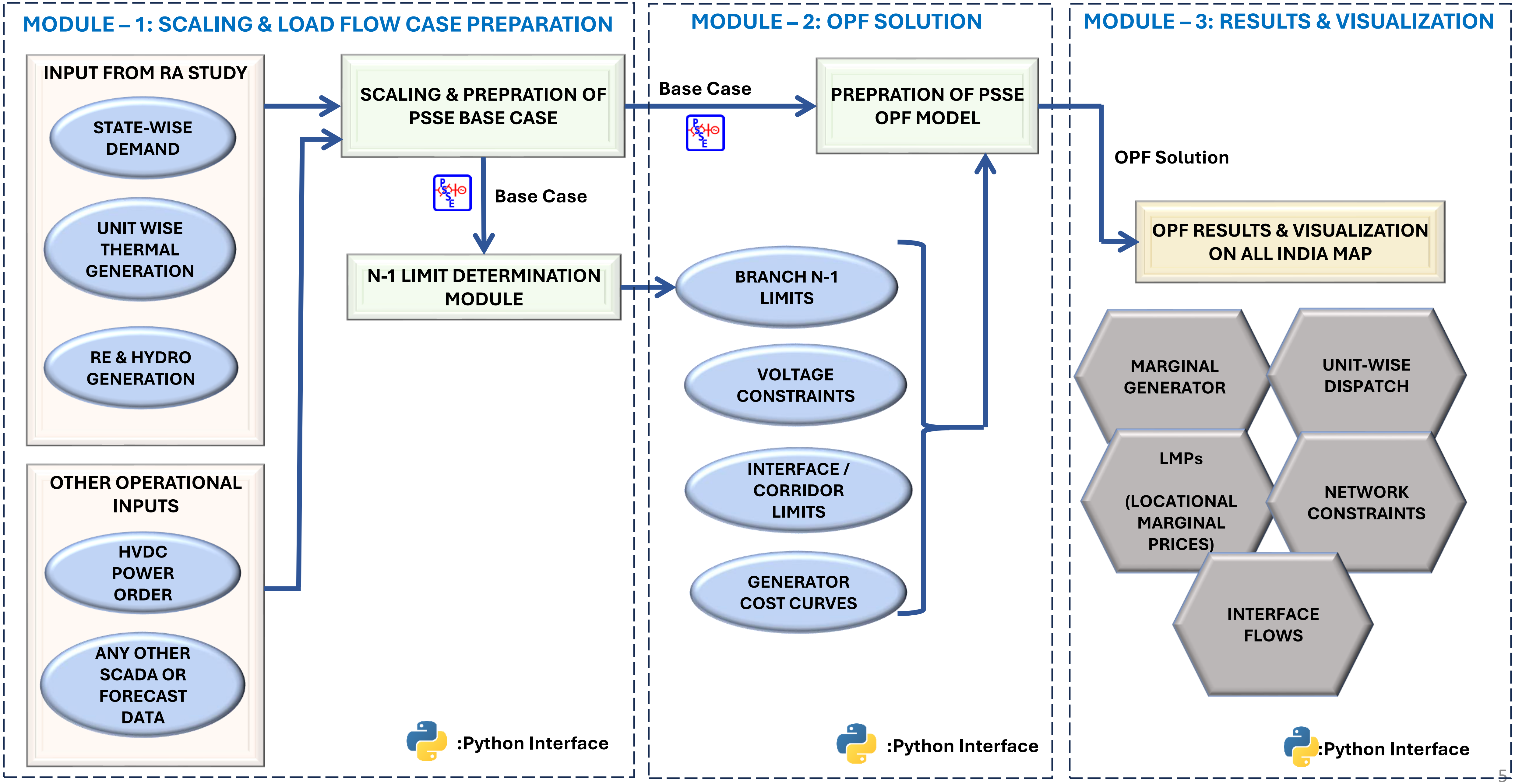
Mapping of Input Data → **All India OPF Study** (Production cost model with detailed transmission) → **OPF Results**

1. Input Data from **Production Cost Model** (may be taken from SCADA also)
   - ➢ Unit-wise thermal (coal, gas, nuclear) generation dispatch
   - ➢ Region-wise solar, wind, and hydro generation
   - ➢ State-wise demand
2. **Variable cost (cost curves)** of thermal generators

1. Complete **Network model with 'N-1' limits**
2. **Constraints:**
   - ➢ Thermal loading limits of 400 kV and above – **Hard Limits**
   - ➢ Voltage limits of 400 kV and above – **Hard Limits**
   - ➢ Inter-regional Import/Export Transfer Capability (NR, SR, NER) – **Hard Limits**

1. **Comparison of results with production cost results (without transmission) before and after the converged OPF solution**
   - ➢ Generation dispatches
   - ➢ Line flow and voltages before and after OPF
   - ➢ Corridor flows
2. **LMP** of all nodes on **all India Map**
3. **Congestion** in the transmission corridor and envisaged market splits

Work Flow - OPTIMA Application



Mapping of Resource Adequacy Study Output Results (Production Cost Model) with PSS/E Network Model

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2. Mapping of Resource Adequacy Study Output Results (Production Cost Model) with PSS/E Network Model

Unit-wise thermal generation scaling

- Total ~**650 thermal (coal, lignite, gas and nuclear)** units have been mapped with respective PSS/E buses and unit ids though python script

State-wise demand scaling

- **State-wise scaling of demand** in the network case through a Python script
- Regional demand is being apportioned to each state as per their contribution to regional demand (depends on the snapshot being scaled)

Solar, Wind and Hydro generation scaling

- Being scaled on a regional basis as per the **dispatch factors/forecasted values** though python script

HVDC Set-point Scaling

- User-defined input – can be scaled in one go through an external excel file and python script

2. Mapping of Resource Adequacy Study Output Results (Production Cost Model) with PSS/E Network Model

Unit-wise thermal generation dispatch and cost scaling

Sample Input Sheet

Sl No	PSSE Bus Number	Unit Number	Thermal Unit (As per RA Database)	Region	Installed Capacity (MW)	Variable cost (Paisa/kWh)	Dispatch
1	352034	T1	ACBIL1	WR	30	109.00	23
2	354028	T1	ACBIL2	WR	135	109.00	125
3	354028	T2	ACBIL3	WR	135	109.00	125
4	352034	T2	ACBIL9	WR	30	459.00	0
5	224002	T1	ADHUNIK1	ER	270	178.00	250
6	224002	T2	ADHUNIK2	ER	270	178.00	250
7	311023	C2	AECO2	WR	120	360.00	111
8	311023	C3	AECO3	WR	121	360.00	112
9	311023	C4	AECO4	WR	121	360.00	112
…	…	…	…	…	…	…	…
…	…	…	…	…	…	…	…
650	434025	T2	YERAMARASTPS2	SR	800	300.00	700

- Total ~**650 thermal (coal, lignite, gas and nuclear)** units have been mapped with respective PSS/E buses and unit ids though python script

2. Mapping of Resource Adequacy Study Output Results (Production Cost Model) with PSS/E Network Model

State-wise demand scaling

Sample Input Sheet

Sl No	State/Entity	PSSE Area Number	Demand Data (MW)
1	PUNJAB	1001	10211
2	HARYANA	1002	9549
3	RAJASTHA	1003	12508
4	DELHI	1004	5122
5	UTTARPRA	1005	20444
6	UTTARAKH	1006	1917
7	HIMACHAL	1007	1227
8	JAMUKASH	1008	1705
9	CHANDIGA	1009	318
...	...	...	...
...	...	...	...
39	TRIPURA	5801	267

- State-wise scaling of demand in the network case through a Python script
- Regional demand is being apportioned to each state as per their contribution to regional demand (depends on the snapshot being scaled)

RE & Hydro Lumped Input

Region	Hydro	Solar	Wind
NR	4500	24000	500
WR	2600	22000	1500
SR	5000	15000	1600
ER	2000	0	0
NER	800	0	0
All India	14900	61000	3600

- RE & Hydro are scaled on a regional basis as per the input though python script
- The generation dispatch input is apportioned in various solar/wind/hydro generators based on installed capacity & dispatch factors.

2. Mapping of Resource Adequacy Study Output Results (Production Cost Model) with PSS/E Network Model

HVDC Pole-wise Power Order

Sample Input Sheet

Sl No	HVDC Name	Maximum Capacity	POWER ORDER	Direction of Operation	Pole -1	Pole - 2	Pole -1 Status	Pole - 2 Status	Rectifier End bus Number	Inverter End bus Number
1	Vindhyachal HVDC	500	400	Forward	VCHL1	VCHL2	Yes	Yes	324030	154038
2	Sasaram HVDC	500	500	Forward	SSRM1		Yes	Yes	214004	214012
3	Bhadrawati HVDC	1000	800	Reverse	BHVT1	BHVT2	Yes	Yes	334008	424014
4	Gazuwaka HVDC	750	600	Reverse	GAZU1	GAZU2	Yes	Yes	414053	414009
5	Talcher-Kolar HVDC	2000	1800	Forward	TKPL1	TKPL2	Yes	Yes	254023	434010
6	Rihand-Dadri HVDC	1500	1000	Forward	RHND1	RHND2	Yes	Yes	154039	154044
7	Balia-Bhiwadi HVDC	1500	1000	Forward	BLIA1	BLIA2	Yes	Yes	154054	134024
8	Chandrapur-Phadge HVDC	1500	1000	Forward	CHND1	CHND2	Yes	Yes	334007	334013
9	Agra - Biswanath Chariali (BNC) - HVDC	3000	1500	Forward	18	19	Yes	Yes	524007	154064
10	Agra - Alipurduar (APD) - HVDC	3000	1500	Forward	APDA1	APDA2	Yes	Yes	264016	154034
11	APL Mundra – Mohindergarh HVDC	2000	1000	Forward	MUND1	MUND2	Yes	Yes	314043	124025
12	Champa – Kurukshetra HVDC Bipole - 1	3000	1500	Forward	CKBP1	CKMP1	Yes	No	354036	124026
13	Champa – Kurukshetra HVDC Bipole - 2	3000	1500	Forward	CKBP2	CKMP2	Yes	No	354045	124026
14	Raigarh- Pugalur HVDC - 1	3000	1500	Forward	RPBP1	RPMP1	Yes	No	354054	444047
15	Raigarh- Pugalur HVDC - 2	3000	1500	Forward	RPBP2	RPMP2	Yes	No	354058	444047

- HVDC power orders, direction of operation are scaled in one go through an external Excel file and python script

Mapping of Data for Optimal Power Flow (OPF) Studies

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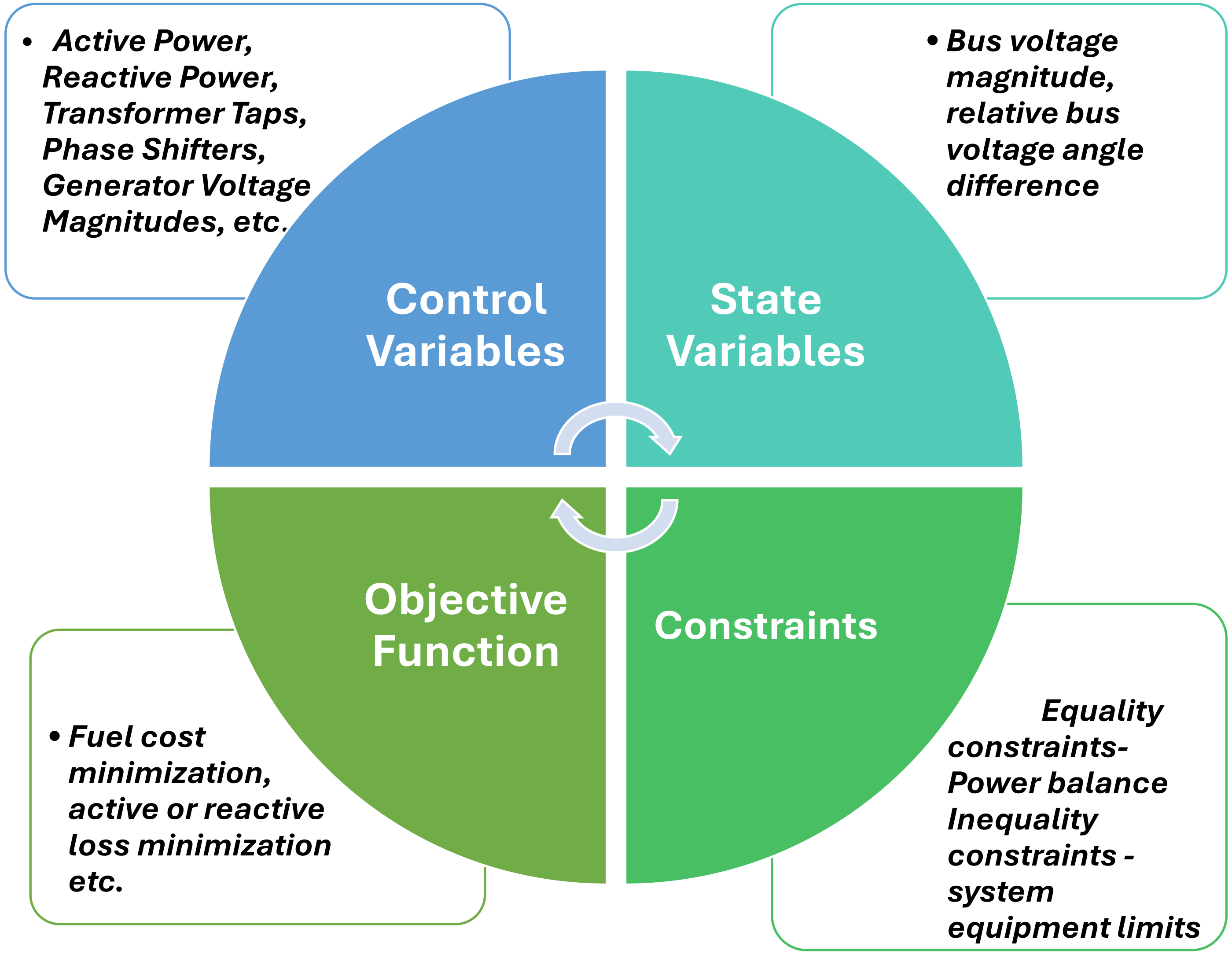
3. Mapping of Data for Optimal Power Flow (OPF) Studies

Components of an OPF Problem



3. Mapping of Data for Optimal Power Flow (OPF) Studies

- PSS®E OPF solves a **nonlinear constrained AC optimisation** problem.

- Objective: Minimise fuel cost (in our case)

- Subject to:

    - **AC power-flow equality constraints** (Active & Reactive power balance equations)
    - **Operating inequality constraints** such as voltage, line flow, interface, generator P & Q limits, and shunt limits.

$$\boldsymbol{min\, f(x)\ \ subject\ to}$$

$$\boldsymbol{h_i(x) = 0\ for\ i = 1, \dots\dots, m}$$
$$\boldsymbol{f_i(x)\ \le 0\ for\ i = 1, \dots\dots, p}$$

$$\boldsymbol{L = f(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} v_i h_i(x)}$$

L = Lagrangian function & v = Lagrangian multiplier

3. Mapping of Data for Optimal Power Flow (OPF) Studies

Treatment of Constraints in PSSE-OPF: Voltage/Branch Flow/Interface Limits

PENALTY LIMIT ON SYSTEM CONTRAINTS

- REPORTING
- SOFT LIMIT
  - **QUADRATIC PENALTIES**

    (Introduces a quadratic penalty term to the objective function to discourage large and infeasible excursions from the mid point of maximum & minimum)
  - **LINEAR PENALTIES**

    (Introduces a linear penalty term to the objective function to discourage large and infeasible excursions from the limits)
- HARD LIMIT
  - Introduces a Barrier function inequality constraint augmenting the Lagrangian in terms of max & min values.

3. Mapping of Data for Optimal Power Flow (OPF) Studies

The cost curves of a total of ~**650 thermal generators** are imported in PSS/E based on **the variable cost** of thermal generators

Currently, **linear cost curves** have been prepared using variable cost data

Interface limits (NR import/export, SR import/export, NER import/export TTC limits) based on presently declared TTC/ATC limits.

PSS®E OPF does not account for HVDC links when evaluating interface limits, nor are HVDC flows reflected in the OPF solution. Therefore, interface limits have been defined considering only AC lines

Computation of **N-1 limits** of each element through automated script and importing to the PSS/E OPF case

Voltage limits imposed on all buses

3. Mapping of Data for Optimal Power Flow (OPF) Studies

Production Cost Curves of few generating stations

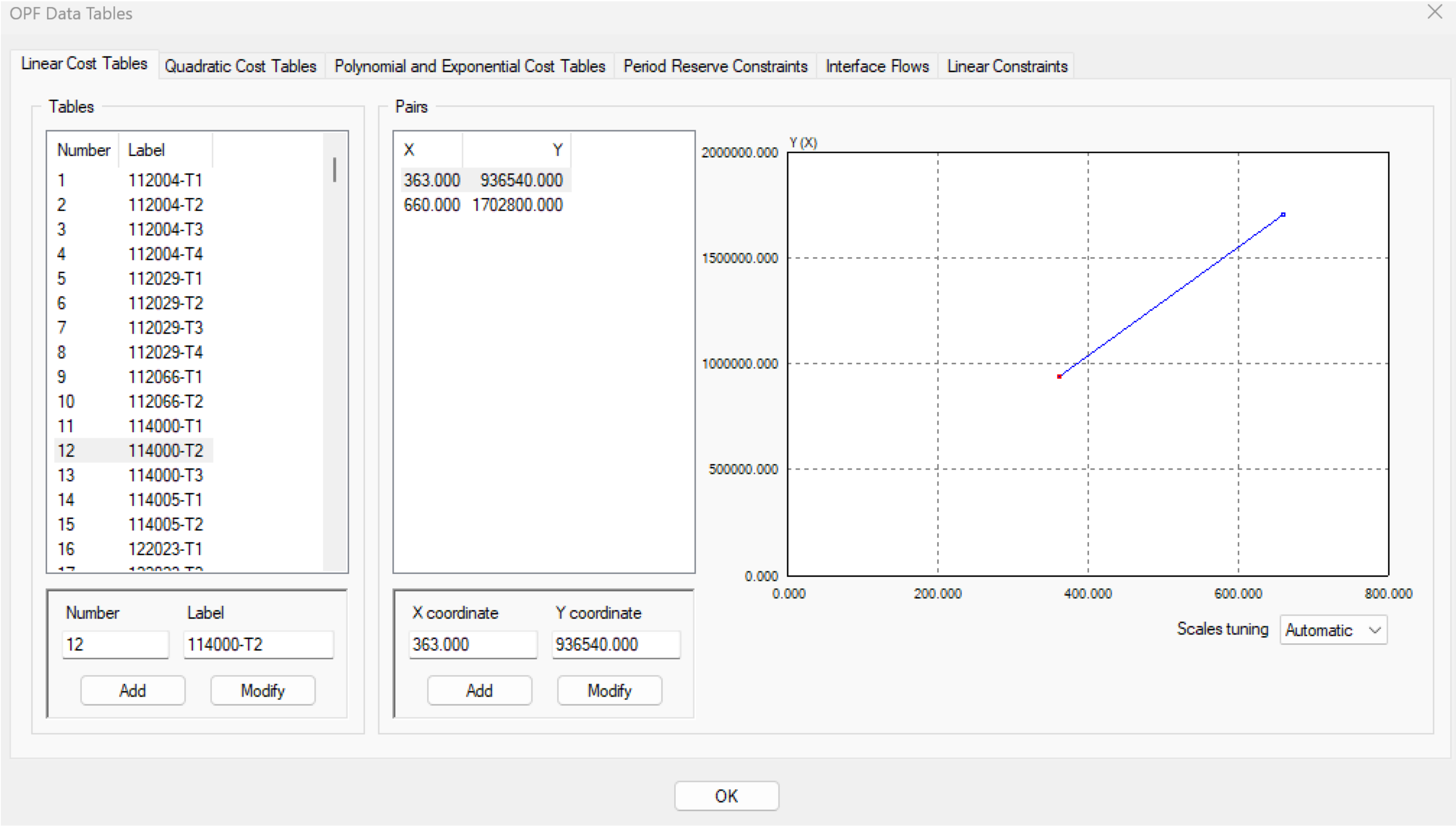
Sample Input Sheets

Sl No	PSSE Bus Number	PSSE Unit ID	Unit Name as Per RA Database	Region	Installed Capacity (MW)	Variable cost (Paisa/kWh)	Point - 1 of Piece-wise Linear Cost Curve (MW)	Point - 1 of Piece-wise Linear Cost Curve (RS/Hr)	Point - 2 of Piece-wise Linear Cost Curve (MW)	Point - 2 of Piece-wise Linear Cost Curve (RS/Hr)
1	352034	T1	ACBIL1	WR	30	109.00	16.5	17985	30	32700
..	..	..	..	..	..	..	..	..	..	..
16	154014	T1	AnparaTPS1	NR	210	216.00	115.5	249480	210	453600
17	154014	T2	AnparaTPS2	NR	210	216.00	115.5	249480	210	453600
18	154014	T3	AnparaTPS3	NR	210	216.00	115.5	249480	210	453600
19	154014	T4	AnparaTPS4	NR	500	183.00	275	503250	500	915000
20	154014	T5	AnparaTPS5	NR	500	183.00	275	503250	500	915000
21	157001	T1	AnparaTPS6	NR	500	186.00	275	511500	500	930000
22	157001	T2	AnparaTPS7	NR	500	186.00	275	511500	500	930000
...	..	..	..	..	..	..	..	..	..	..
39	264004	T1	BAKRESHWAR1	ER	210	252.00	115.5	291060	210	529200
40	264004	T2	BAKRESHWAR2	ER	210	252.00	115.5	291060	210	529200
41	262010	T3	BAKRESHWAR3	ER	210	252.00	115.5	291060	210	529200
42	262010	T4	BAKRESHWAR4	ER	210	252.00	115.5	291060	210	529200
43	262010	T5	BAKRESHWAR5	ER	210	252.00	115.5	291060	210	529200
..	..	..	..	..	..	..	..	..	..	..
649	434025	T1	YERAMARASTPS1	SR	800	300	440	1320000	800	2400000
650	434025	T2	YERAMARASTPS2	SR	800	300	440	1320000	800	2400000

Piece-wise linear cost curve: Helps obtain the relative fuel cost for dispatching a participating generator unit at a certain active power dispatch level

3. Mapping of Data for Optimal Power Flow (OPF) Studies

Production Cost Curves of few generating stations

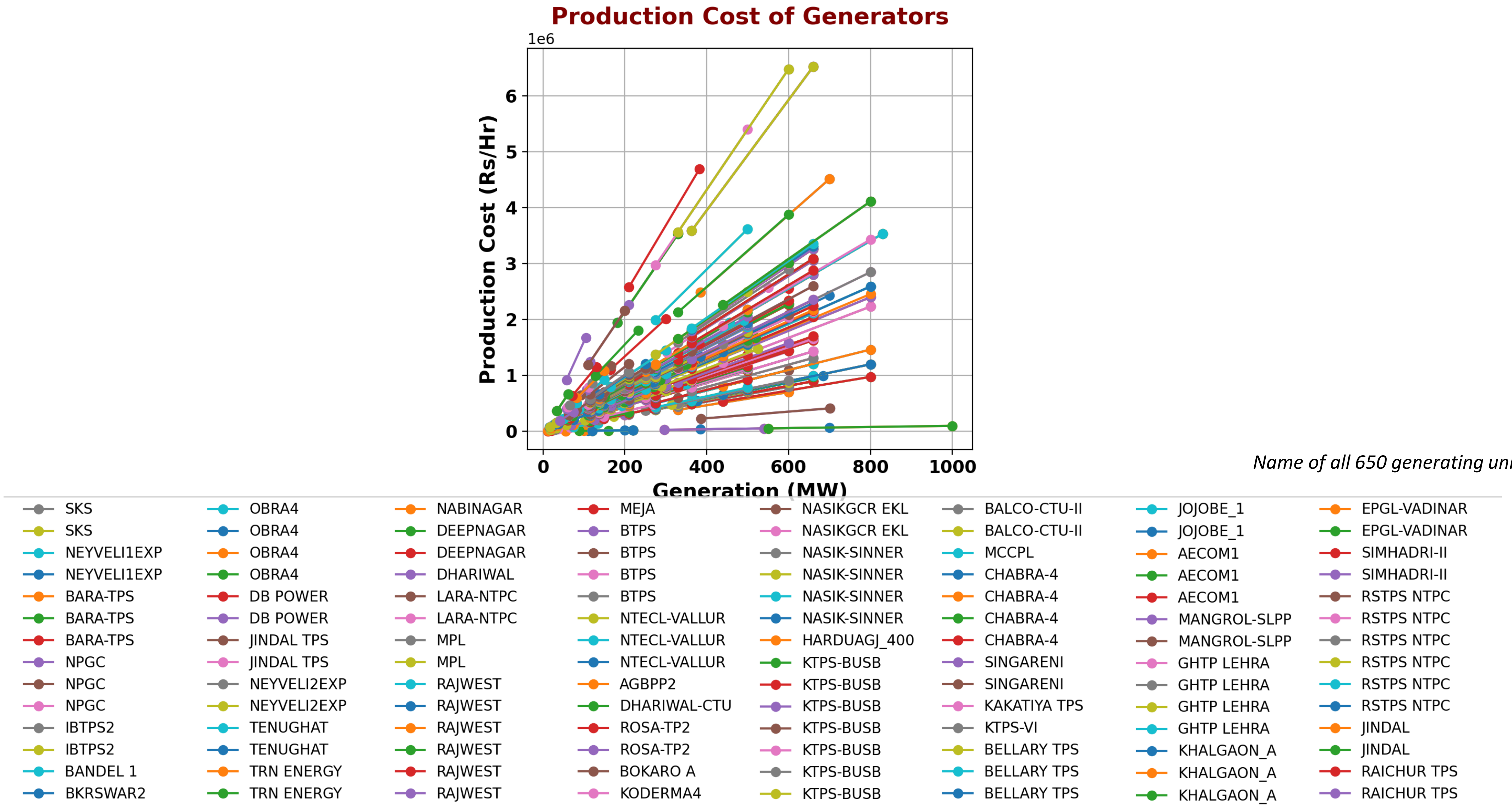


Sample PSSE Snapshot

Piece-wise linear cost curve: Helps obtain the relative fuel cost for dispatching a participating generator unit at a certain active power dispatch level

3. Mapping of Data for Optimal Power Flow (OPF) Studies

Production Cost Curves of few generating stations



Name of all 650 generating units may not be visible

3. Mapping of Data for Optimal Power Flow (OPF) Studies

Voltage Limits: Stricter band on 400 & 765 kV buses, slightly relaxed at 220 kV and below level

Sample PSS/E Snapshot

Bus Number	Section Number	Substation Number	Bus Name	Normal OPF Vmax (pu)	Normal OPF Vmin (pu)	Adjust Normal V Limits	Emergency OPF Vmax (pu)	Emergency OPF Vmin (pu)	Limit Type	Soft Limit Penalty
117000			MOGA-PG 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
127000			BHIWN-PG_7 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
127001			MERUT_NARELA 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
127002			BHIW_NARELA 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
137000			PHAGI 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
137001			ANTA 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
137002			CHITTOR_765 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
137003			AJMER_765 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
137004			BIKANER-PG 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
137005			BHADLA-PG 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
137006			BHADLA II-PG 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
137007			FATEHGARH-I 765.00	2.0000	0.2000	No change	2.0000	0.2000	Hard limit	1.0000
137008			FATEHGARH-II 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
137009			KHETRI 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
137010			SIKAR_2 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000
137011			FATEHGARH-3 765.00	1.1500	0.8500	No change	1.3000	0.7000	Hard limit	1.0000

- **Normal voltage limits considered for all voltage level – 1.15 p.u. to 0.85 p.u. (Hard Limits)**

3. Mapping of Data for Optimal Power Flow (OPF) Studies

Branch Constraints: N-1 Limits of 400 & 765 kV elements (ICTs & transmission lines)

From Bus Number	From Bus Name	To Bus Number	To Bus Name	Ckt ID	Rate2 (MVA)	N-1 Limit
157018	ALIGARH	154087	ALIGARH (PG)	1	1500	862
157018	ALIGARH	154087	ALIGARH (PG)	2	1500	862
157018	ALIGARH	147000	JHATI-PG	B1	4200	2359
157018	ALIGARH	157020	ALGRH_AGRA	H1	4200	3182
157018	ALIGARH	157019	ALGRH_GNOIDA	H1	3880	3255
157018	ALIGARH	157017	ORAI	H1	4200	3347
157018	ALIGARH	157017	ORAI	H2	4200	3347
157019	ALGRH_GNOIDA	157011	GR.NOIDA	B1	3880	3596
157020	ALGRH_AGRA	157007	AGRA-PG	B1	4200	3787
157021	ALGRH_KANPUR	157018	ALIGARH	H1	4200	3387
157023	GHATAMPUR	157025	RAMPUR_PRSTL	B1	4200	1970
157023	GHATAMPUR	157013	FATEHBD_AGRA	B1	3879.7	2059
157024	KOT-MER-FSC2	167000	KOTESHWAR	B2	4200	2352
157025	RAMPUR_PRSTL	154109	RAMPUR_PRSTL	1	1500	898
…	…	…	…	…	…	…
…	…	…	…	…	…	…
157026	MEERUT_PMSTL	154111	MEERUT_PMSTL	1	1500	922
157026	MEERUT_PMSTL	154111	MEERUT_PMSTL	2	1500	930
157028	JAWAHARPUR	157011	GR.NOIDA	B1	4200	2285

Sample Input Sheets

Sample PSSE Snapshot

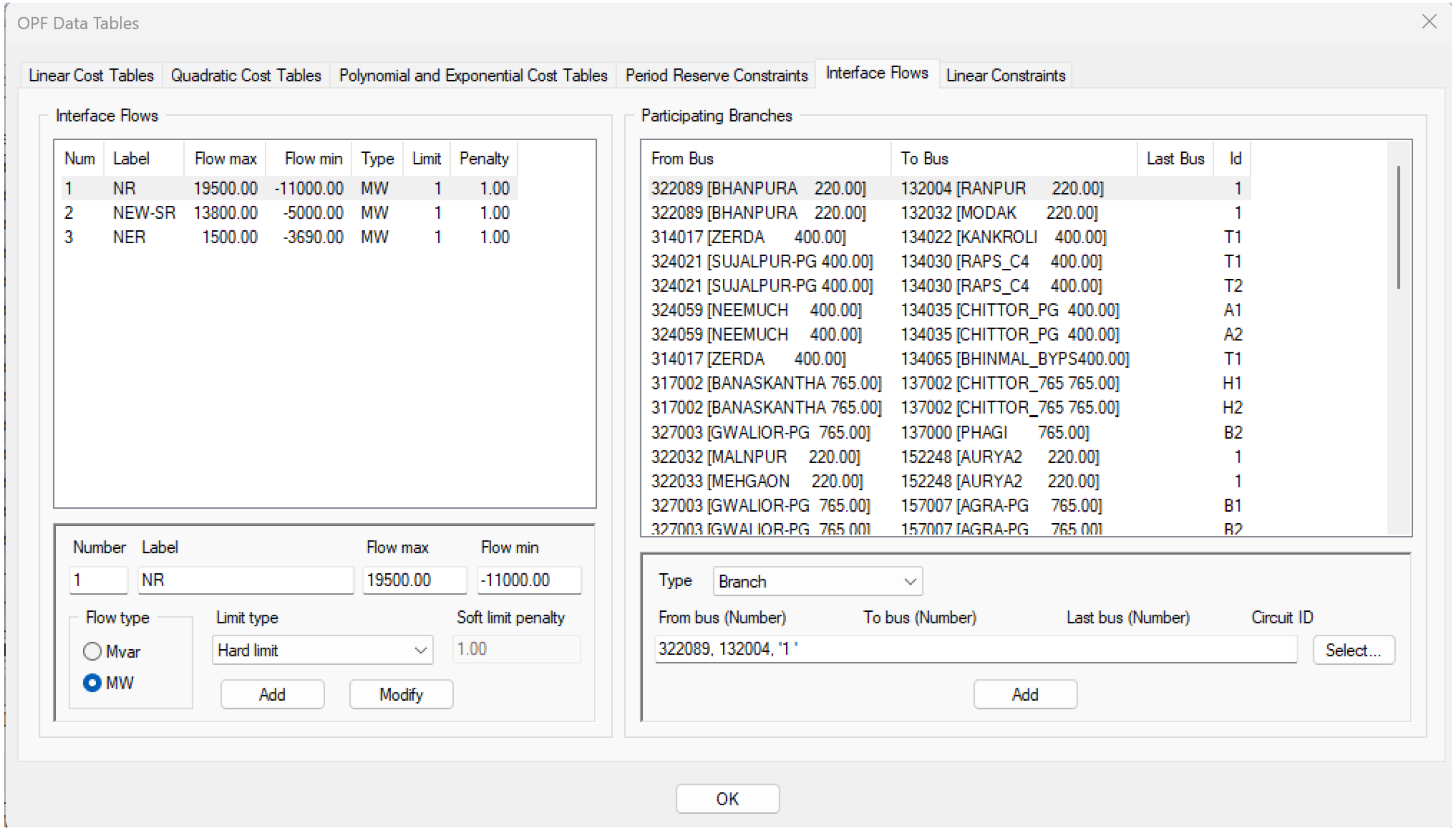
From Bus Number	From Bus Name	To Bus Number	To Bus Name	Id	Flow Id	Normal Flow Limit Max	Normal Flow Limit Min	Initialize Rate Limits	Emergency Flow Limit Max	Emergency Flow Limit Min	Flow Type	Limit Type	Soft Limit Penalty
114007	MOGA 400.00	117000	MOGA-PG 765.00	1	1	872.20	-872.20	None	872.20	-872.20	MVA	Reporting	1.00
114007	MOGA 400.00	117000	MOGA-PG 765.00	2	1	872.20	-872.20	None	872.20	-872.20	MVA	Reporting	1.00
117000	MOGA-PG 765.00	127000	BHIWN-PG_7 765.00	B1	1	3165.30	-3165.30	None	3165.30	-3165.30	MVA	Reporting	1.00
117000	MOGA-PG 765.00	137004	BIKANER-PG 765.00	H1	1	3108.10	-3108.10	None	3108.10	-3108.10	MVA	Reporting	1.00
117000	MOGA-PG 765.00	137004	BIKANER-PG 765.00	H2	1	3108.10	-3108.10	None	3108.10	-3108.10	MVA	Reporting	1.00
117000	MOGA-PG 765.00	157008	MERUT-PG 765.00	B1	1	3312.90	-3312.90	None	3312.90	-3312.90	MVA	Reporting	1.00
124021	BHIWANI-PG 400.00	127000	BHIWN-PG_7 765.00	2	1	759.10	-759.10	None	759.10	-759.10	MVA	Reporting	1.00
124021	BHIWANI-PG 400.00	127000	BHIWN-PG_7 765.00	3	1	759.10	-759.10	None	759.10	-759.10	MVA	Reporting	1.00
124031	BHIWANI-PG2 400.00	127000	BHIWN-PG_7 765.00	1	1	636.40	-636.40	None	636.40	-636.40	MVA	Reporting	1.00
127000	BHIWN-PG_7 765.00	127002	BHIW_NARELA 765.00	B1	1	3142.30	-3142.30	None	3142.30	-3142.30	MVA	Reporting	1.00
127000	BHIWN-PG_7 765.00	137000	PHAGI 765.00	B1	1	3275.20	-3275.20	None	3275.20	-3275.20	MVA	Reporting	1.00
127000	BHIWN-PG_7 765.00	137000	PHAGI 765.00	B2	1	3275.20	-3275.20	None	3275.20	-3275.20	MVA	Reporting	1.00
127000	BHIWN-PG_7 765.00	147000	JHATI-PG 765.00	B1	1	3018.30	-3018.30	None	3018.30	-3018.30	MVA	Reporting	1.00
127001	MERUT_NARELA 765.00	147001	NARELA 765.00	H1	1	3646.20	-3646.20	None	3646.20	-3646.20	MVA	Reporting	1.00
127001	MERUT_NARELA 765.00	157008	MERUT-PG 765.00	B1	1	3379.00	-3379.00	None	3379.00	-3379.00	MVA	Reporting	1.00
127002	BHIW_NARELA 765.00	147001	NARELA 765.00	H1	1	2723.70	-2723.70	None	2723.70	-2723.70	MVA	Reporting	1.00

Bus Voltage | Bus Voltage / Sbs | Adj. Bus Shunt | Adj. Bus Shunt / Sbs | Adj. Branch Reac. | Adj. Branch Reac. / Sbs | **Branch Flow** | Branch Flow / Sbs | Adj. Bus Load | Adj. Bus Load / Sbs | Adj. Load Table

- **N-1 limits computed for each element have been imported in the OPF case.**
- **Limits considered for 765 kV and 400 kV elements in the final run.**

3. Mapping of Data for Optimal Power Flow (OPF) Studies

Interface Limits: ATC limits of SR Import/Export, NR Import/Export & NER Import/Export



Sample PSSE Snapshot

OPF Results

(1. High SR Import Scenario)

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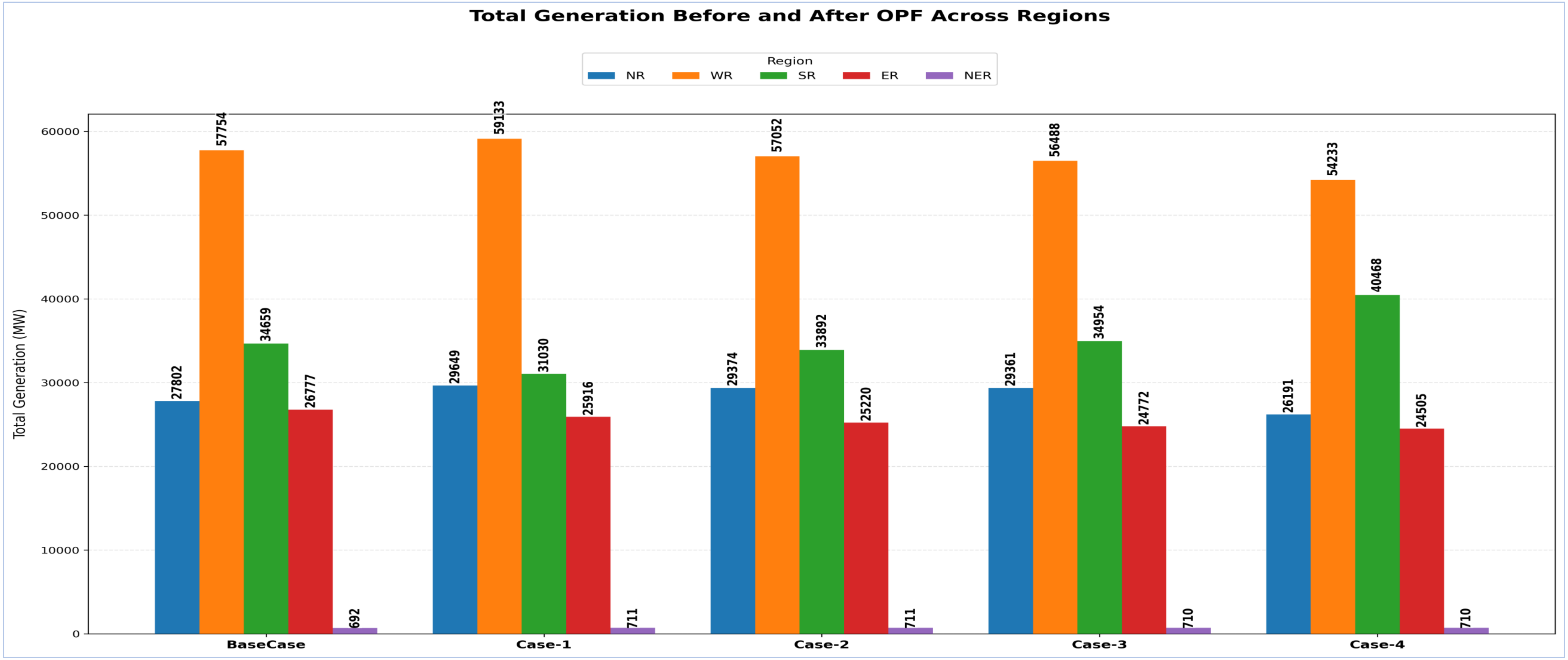
5. OPF Results

SR Import Scenario

Study Cases	Description	Interface Constraints (ATC/TTC Limits)	Branch Flow (N-1) Limit Constraints	Voltage Limit Constraints
Case – 1	Voltage limits imposed + No network constraints	❌	❌	✅
Case – 2	Voltage limits + Interface limits imposed	✅	❌	✅
Case – 3	Voltage limits + Branch flow limits imposed	❌	✅	✅
Case – 4	Voltage limits + Branch flow limits + Interface limits imposed (HVDC Raigarh - Pugalur Under Outage, ~6000 MW TTC Reduction)	✅	✅	✅

5. OPF Results

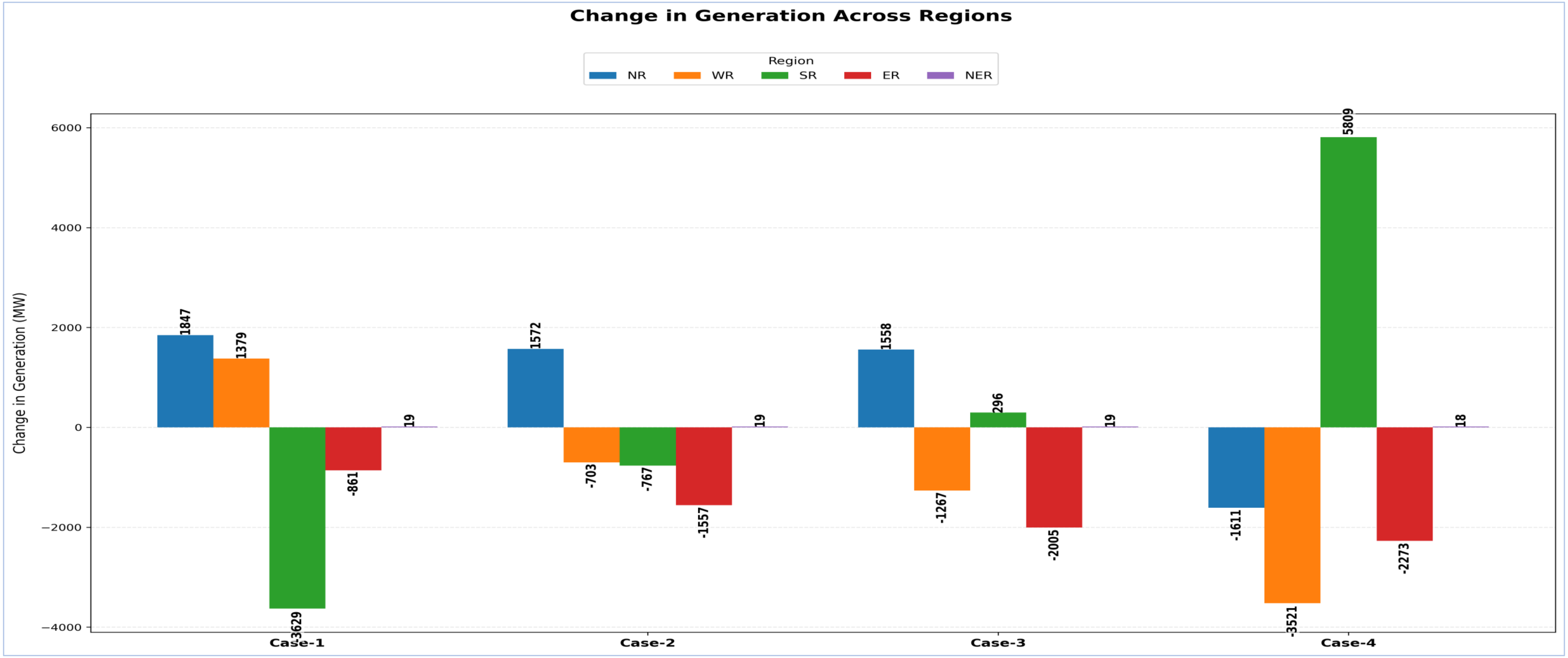
SR Import Scenario



Case – 1 (Copper Plate)	Case – 2 (Interface constraint imposed & binding)	Case – 3 (Branch limit constraint imposed & binding)	Case – 4 (Both interface & branch limit constraints imposed & binding)

5. OPF Results

SR Import Scenario



Case – 1 (Copper Plate)	Case – 2 (Interface constraint imposed & binding)	Case – 3 (Branch limit constraint imposed & binding)	Case – 4 (Both interface & branch limit constraints imposed & binding)

5. OPF Results

SR Import Scenario

Corridor/Transmission Constraint	TTC/Safe Operating Limit (MW)	Power Flows in MW Across Various Cases (MW)				
		Base Case	Case – 1 (Copper Plate)	Case – 2 (Interface constraint imposed & binding)	Case – 3 (Branch limit constraint imposed & binding)	Case – 4 (Both interface & branch limit constraints imposed & binding)
ER-SR	**6500**	6600	7302	6530	6280	6250
WR-SR	**15800**	16850	19192	16120	15650	10400
SR Import	**22300**	**23450**	**26546**	22650	21950	16650
765 kV Angul - Srikakulam D/C (Each Ckt)	**1850**	**1908**	**2283**	**1963**	1842	1825
765 kV Warora - Warangal D/C (Each Ckt)	**1900**	**1960**	**2357**	**2007**	1841	1839
765 kV Wardha - Nizamabad D/C (Each Ckt)	**1850**	**1940**	**2284**	**1950**	1837	1770

- **Base case:** SR import violations were there under the base case
- **Case – 1:** With copper plate model (no transmission constraint imposed), the SR import flow violated the TTC limits
- **Case – 2 :** With the interface limits, the corridor flows were restricted but the line loadings violated the N-1 limits
- **Case – 3 :** With the branch limits, the line loadings violations were addressed
- **Case - 4:** Under transmission outage and binding interface & branch constraints, major change in generation dispatch, corridor flows and LMPs

5. OPF Results

SR Import Scenario

Region	Case - 1		Case - 2		Case - 3		Case - 4	
	Marginal Generator	LMP of Marginal Generator Bus	Marginal Generator	LMP of Marginal Generator Bus	Marginal Generator	LMP of Marginal Generator Bus	Marginal Generator	LMP of Marginal Generator Bus
SR	-	-	Hinduja (3.22 Rs/KWh)	3.21 Rs./KWh	SDSTPS (3.24 Rs/KWh)	3.24 Rs/KWh	Ramagundam (3.83 Rs/KWh)	3.83 Rs/KWh
Rest of India	Mahan (2.40 Rs/KWh)	2.40 Rs/KWh	Anpara (2.16 Rs/KWh)	2.14 Rs./KWh	Anpara (2.16 Rs/KWh)	2.10 Rs/KWh	Singrauli (1.49 Rs/KWh)	1.49 Rs/KWh

Region	Average LMP (Rs/KWh)			
	Case - 1	Case - 2	Case - 3	Case - 4
NR	2.40	2.21	2.06	1.56
WR	2.57	2.36	2.34	1.81
SR	2.74	3.38	3.35	4.10
ER	2.47	2.27	2.19	1.63
NER	2.44	2.25	2.15	1.61

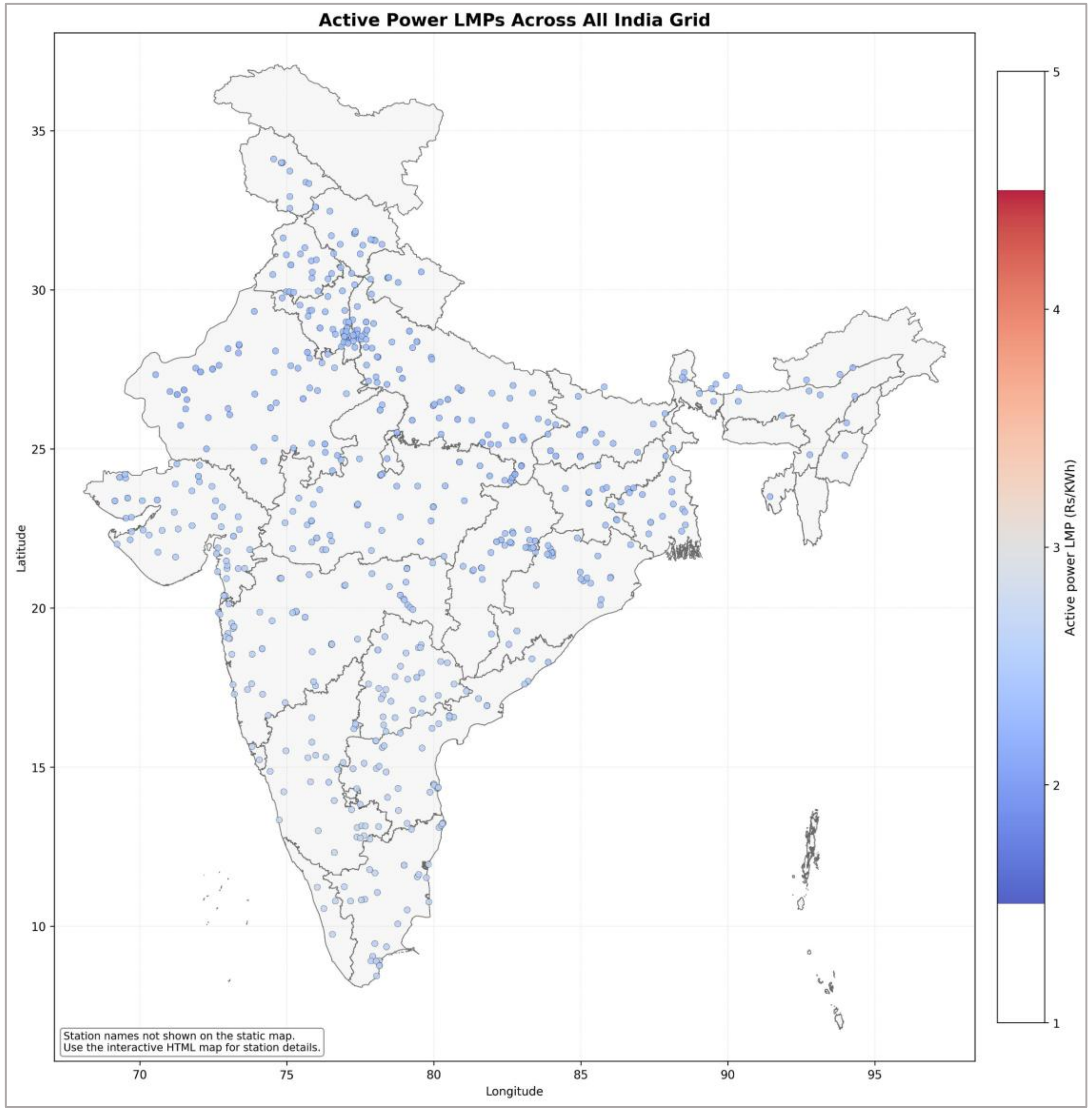
LMPs on all India Map:

Static Maps

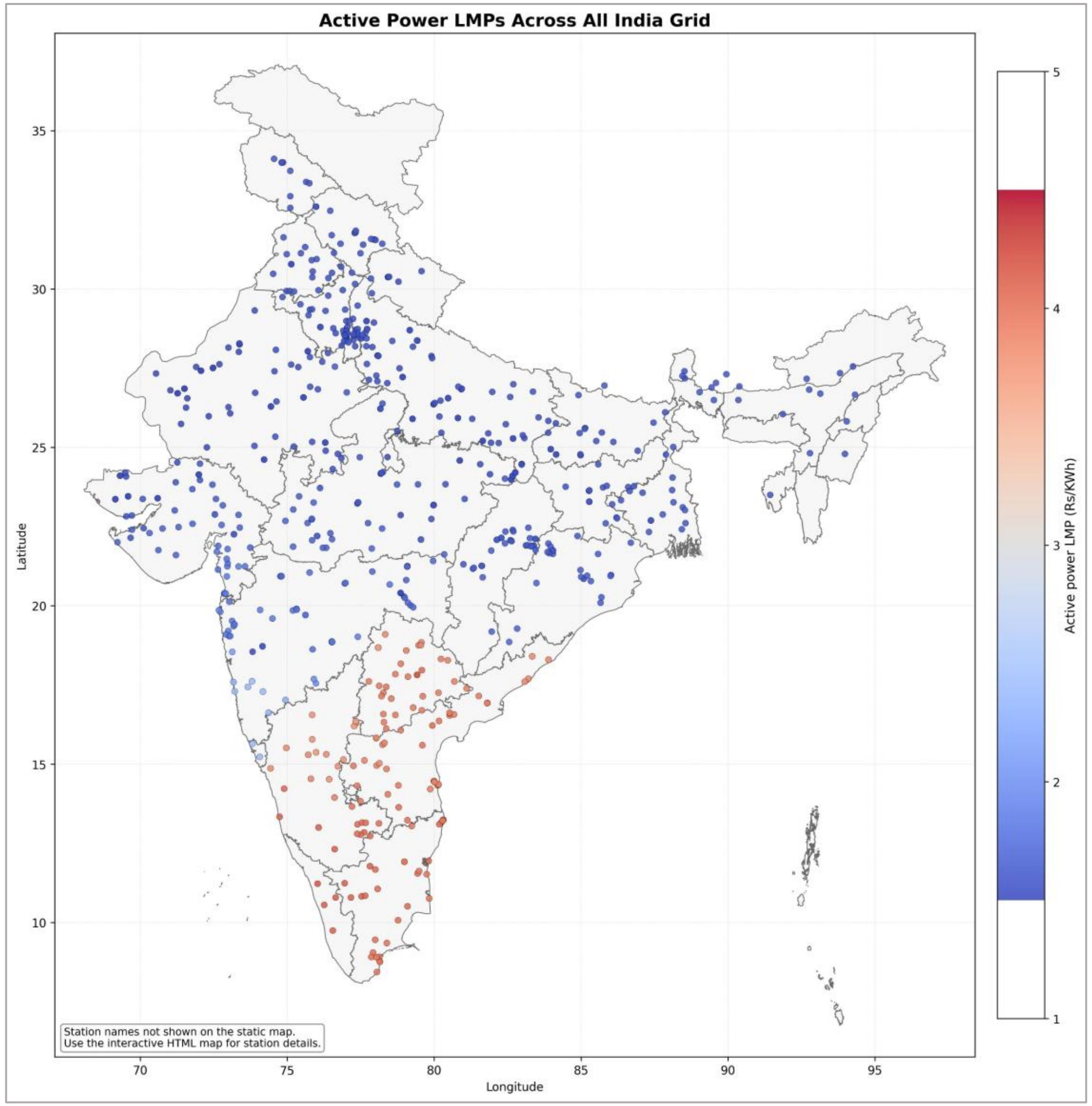
Interactive Maps

- **Base case:** SR import violations were there under the base case
- **Case – 1:** With copper plate model (no transmission constraint imposed), the SR import flow violated the TTC limits
- **Case – 2 :** With the interface limits, the corridor flows were restricted but the line loadings violated the N-1 limits
- **Case – 3 :** With the branch limits, the line loadings violations were addressed
- **Case - 4:** Under transmission outage and binding interface & branch constraints, major change in generation dispatch, corridor flows and LMPs

5. OPF Results
SR Import Scenario



Case – 1: Copper plate model



Case – 4: Both interface & branch limit constraints imposed & binding

5. OPF Results

SR Import Scenario

Study Cases	Description	Link to Interactive Maps
Case – 1	Voltage limits imposed + No network constraints	
Case – 2	Voltage limits + Interface limits imposed	[Click here to go to the link]()
Case – 3	Voltage limits + Branch flow limits imposed	
Case – 4	Voltage limits + Branch flow limits + Interface limits imposed (HVDC Raigarh - Pugalur Under Outage, ~6000 MW TTC Reduction)	**Scenario to be chosen from the dropdown**

5. OPF Results

SR Import Scenario

Case-1 | SR Import Summary OPF										
Top 10 Cheapest Generators (Sorted by Variable Cost - Ascending)										
Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
324028	JAYPEE NIGRI	T1	700	385	59	638	700	62	0	315
324028	JAYPEE NIGRI	T2	700	385	59	640	700	60	0	315
352034	ACBIL	T1	30	17	109	17	30	13	0	13
354028	ACBIL	T1	135	74	109	109	135	25	0	61
354028	ACBIL	T2	135	74	109	106	135	29	0	61
254021	JITPL	T1	600	330	117	368	600	232	0	270
254021	JITPL	T2	600	330	117	368	600	232	0	270
257002	DARLIPALLI	T1	800	440	122	752	800	48	0	360
257002	DARLIPALLI	T2	800	440	122	752	800	48	0	360
354032	JPL STAGE-II	T1	600	330	133	548	600	52	0	270
Top 10 Costliest Generators (Sorted by Variable Cost - Descending)										
Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
332199	ABHIJEET220	T1	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T2	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T3	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T4	62	34	1,083	35	34	-1	28	0
312027	UKAI-T	T4	200	110	1,080	119	110	-9	90	0
312027	UKAI-T	T5	210	115	1,080	116	116	0	94	0
434021	UDUPI PCL	T1	600	330	1,080	437	330	-107	270	0
434021	UDUPI PCL	T2	600	330	1,080	478	330	-148	270	0
312055	APL MUNDRA	T1	330	182	1,070	182	182	0	148	0
312055	APL MUNDRA	T2	330	182	1,070	182	182	0	148	0

Gas, Nuclear & Off-bar thermal units not included in this table

5. OPF Results

SR Import Scenario

Case-2 | SR Import Summary OPF

Top 10 Cheapest Generators (Sorted by Variable Cost - Ascending)

Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
324028	JAYPEE NIGRI	T1	700	385	59	638	700	62	0	315
324028	JAYPEE NIGRI	T2	700	385	59	640	700	60	0	315
352034	ACBIL	T1	30	17	109	17	30	13	0	13
354028	ACBIL	T1	135	74	109	109	135	25	0	61
354028	ACBIL	T2	135	74	109	106	135	29	0	61
254021	JITPL	T1	600	330	117	368	600	232	0	270
254021	JITPL	T2	600	330	117	368	600	232	0	270
257002	DARLIPALLI	T1	800	440	122	752	800	48	0	360
257002	DARLIPALLI	T2	800	440	122	752	800	48	0	360
354032	JPL STAGE-II	T1	600	330	133	548	600	52	0	270

Top 10 Costliest Generators (Sorted by Variable Cost - Descending)

Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
332199	ABHIJEET220	T1	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T2	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T3	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T4	62	34	1,083	35	34	-1	28	0
312027	UKAI-T	T4	200	110	1,080	119	110	-9	90	0
312027	UKAI-T	T5	210	115	1,080	116	116	0	94	0
434021	UDUPI PCL	T1	600	330	1,080	437	330	-107	270	0
434021	UDUPI PCL	T2	600	330	1,080	478	330	-148	270	0
312055	APL MUNDRA	T1	330	182	1,070	182	182	0	148	0
312055	APL MUNDRA	T2	330	182	1,070	182	182	0	148	0

Gas, Nuclear & Off-bar thermal units not included in this table

5. OPF Results

SR Import Scenario

Case-3 | SR Import Summary OPF

Top 10 Cheapest Generators (Sorted by Variable Cost - Ascending)

Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
324028	JAYPEE NIGRI	T1	700	385	59	638	700	62	0	315
324028	JAYPEE NIGRI	T2	700	385	59	640	700	60	0	315
352034	ACBIL	T1	30	17	109	17	30	13	0	13
354028	ACBIL	T1	135	74	109	109	135	25	0	61
354028	ACBIL	T2	135	74	109	106	135	29	0	61
254021	JITPL	T1	600	330	117	368	600	232	0	270
254021	JITPL	T2	600	330	117	368	600	232	0	270
257002	DARLIPALLI	T1	800	440	122	752	800	48	0	360
257002	DARLIPALLI	T2	800	440	122	752	800	48	0	360
354032	JPL STAGE-II	T1	600	330	133	548	600	52	0	270

Top 10 Costliest Generators (Sorted by Variable Cost - Descending)

Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
332199	ABHIJEET220	T1	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T2	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T3	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T4	62	34	1,083	35	34	-1	28	0
312027	UKAI-T	T4	200	110	1,080	119	110	-9	90	0
312027	UKAI-T	T5	210	115	1,080	116	116	0	94	0
434021	UDUPI PCL	T1	600	330	1,080	437	330	-107	270	0
434021	UDUPI PCL	T2	600	330	1,080	478	330	-148	270	0
312055	APL MUNDRA	T1	330	182	1,070	182	182	0	148	0
312055	APL MUNDRA	T2	330	182	1,070	182	182	0	148	0

Gas, Nuclear & Off-bar thermal units not included in this table

5. OPF Results

SR Import Scenario

Case-4 | SR Import Summary OPF

Top 10 Cheapest Generators (Sorted by Variable Cost - Ascending)

Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
324028	JAYPEE NIGRI	T1	700	385	59	638	700	62	0	315
324028	JAYPEE NIGRI	T2	700	385	59	640	700	60	0	315
352034	ACBIL	T1	30	17	109	17	30	13	0	13
354028	ACBIL	T1	135	74	109	109	135	25	0	61
354028	ACBIL	T2	135	74	109	106	135	29	0	61
254021	JITPL	T1	600	330	117	368	600	232	0	270
254021	JITPL	T2	600	330	117	368	600	232	0	270
257002	DARLIPALLI	T1	800	440	122	752	800	48	0	360
257002	DARLIPALLI	T2	800	440	122	752	800	48	0	360
354032	JPL STAGE-II	T1	600	330	133	548	600	52	0	270

Top 10 Costliest Generators (Sorted by Variable Cost - Descending)

Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
332199	ABHIJEET220	T1	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T2	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T3	62	34	1,083	35	34	-1	28	0
332199	ABHIJEET220	T4	62	34	1,083	35	34	-1	28	0
312027	UKAI-T	T4	200	110	1,080	119	110	-9	90	0
312027	UKAI-T	T5	210	115	1,080	116	116	0	94	0
434021	UDUPI PCL	T1	600	330	1,080	437	330	-107	270	0
434021	UDUPI PCL	T2	600	330	1,080	478	330	-148	270	0
312055	APL MUNDRA	T1	330	182	1,070	182	182	0	148	0
312055	APL MUNDRA	T2	330	182	1,070	182	182	0	148	0

Gas, Nuclear & Off-bar thermal units not included in this table

OPF Results

(2. High NR Import Scenario)

Back to Contents

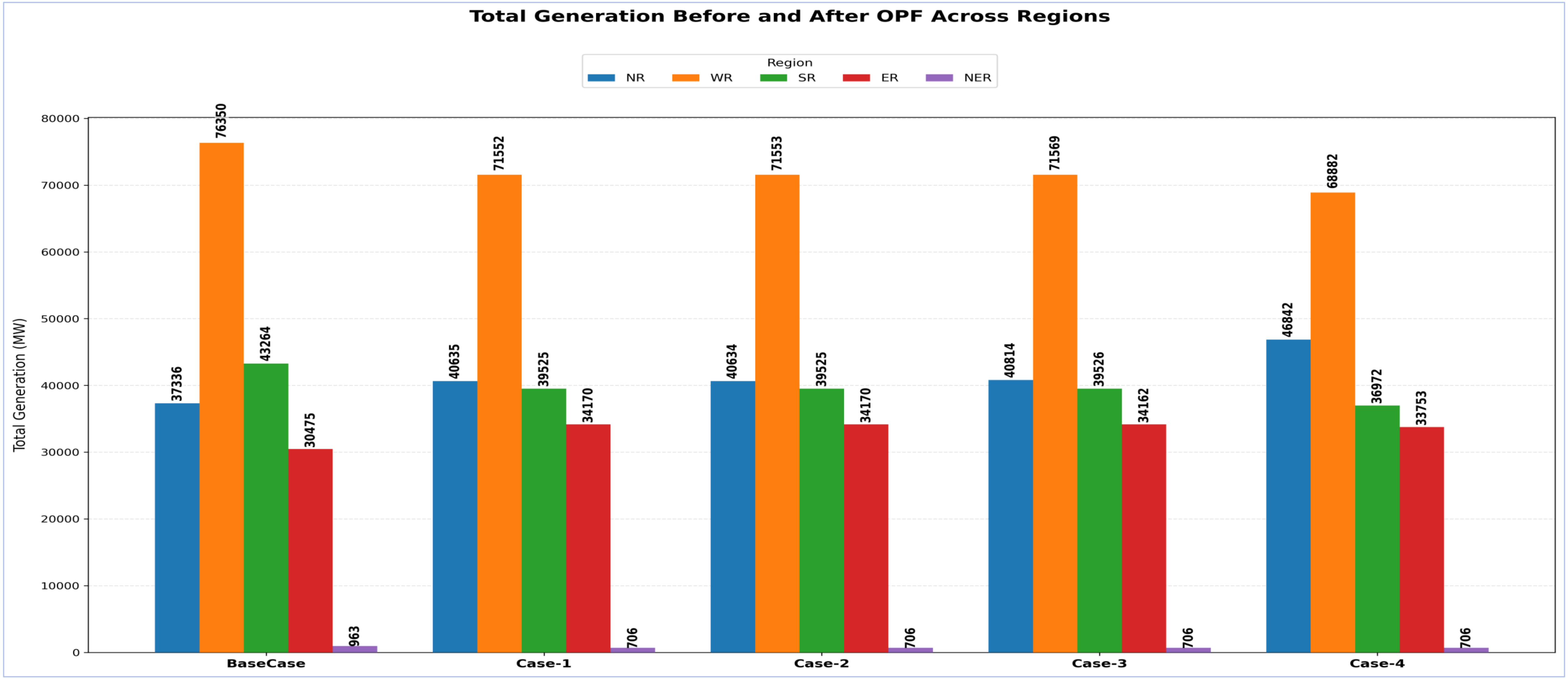
5. OPF Results

NR Import Scenario

Study Cases	Description	Interface Constraints (ATC/TTC Limits)	Branch Flow (N-1) Limit Constraints	Voltage Limit Constraints
Case – 1	Voltage limits imposed + No network constraints	❌	❌	✅
Case – 2	Voltage limits + Interface limits imposed	✅	❌	✅
Case – 3	Voltage limits + Branch flow limits imposed	❌	✅	✅
Case – 4	Voltage limits + Branch flow limits + interface limits imposed (HVDC Champa – Kurukshetra Under Outage, ~6000 MW TTC Reduction)	✅	✅	✅

5. OPF Results

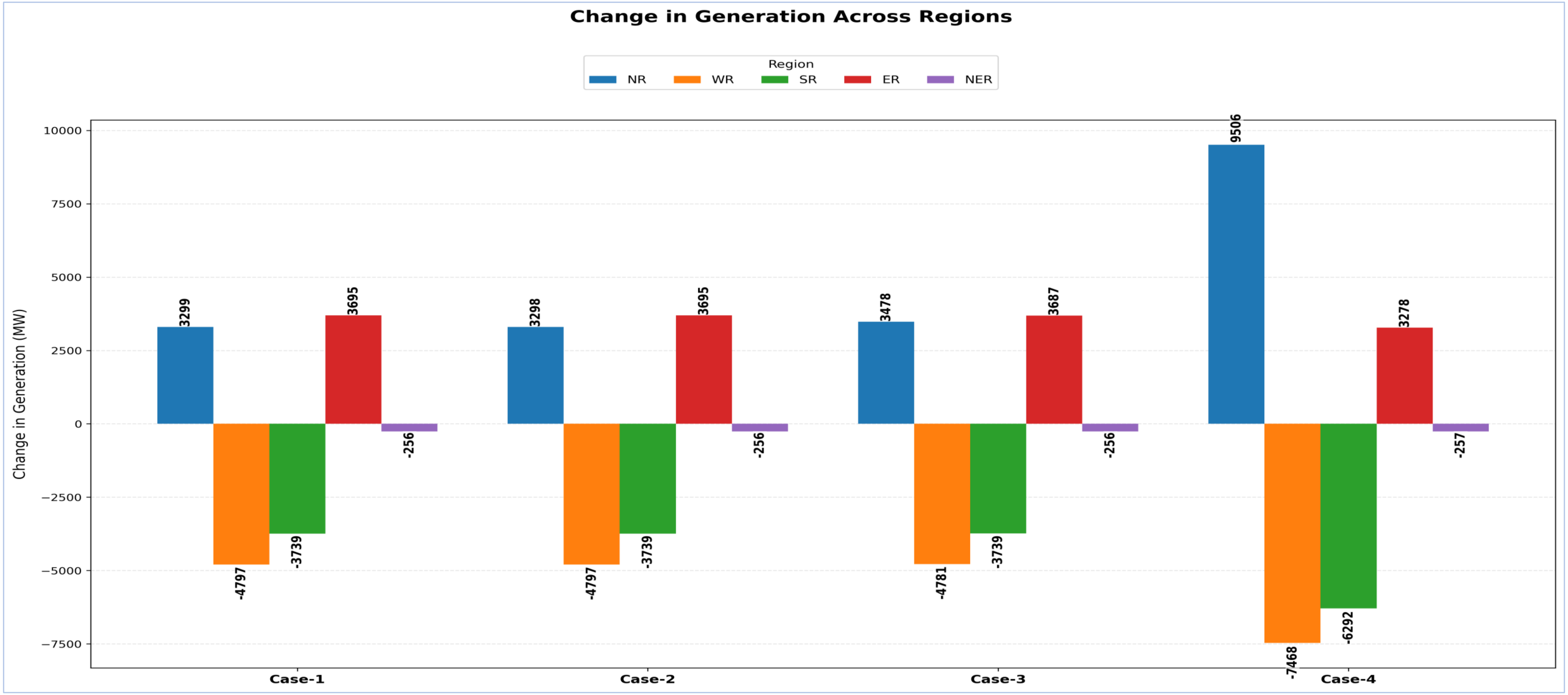
NR Import Scenario



Case – 1 (Copper Plate)	Case – 2 (Interface constraint imposed but not binding)	Case – 3 (Branch limit constraint imposed but not binding)	Case – 4 (Both interface & branch limit constraints imposed & binding)

5. OPF Results

NR Import Scenario



Case – 1 (Copper Plate)	Case – 2 (Interface constraint imposed but not binding)	Case – 3 (Branch limit constraint imposed & binding)	Case – 4 (Both interface & branch limit constraints imposed & binding)

5. OPF Results

NR Import Scenario

Corridor/Transmission Constraint	TTC/Safe Operating Limit (MW)	Power Flows in MW Across Various Cases (MW)				
		Base Case	Case – 1 (Copper Plate)	Case – 2 (Interface constraint imposed but not binding)	Case – 3 (Branch limit constraint imposed & binding)	Case – 4 (Both interface & branch limit constraints imposed & binding)
ER-NR	6300	4200	4703	4702	4737	5029
WR-NR	22350	22950	18386	18838	19082	12540
NR Import	25700	27150	23089	23540	23819	17569
765 kV Vindhyachal - Varanasi D/C (Each Ckt)	1850	1966	1530	1526	1530	1571

- **Base case:** NR import violations were there under the base case
- **Case – 1:** With copper plate model (no transmission constraint imposed), the NR import flow got within the TTC limits
- **Case – 2 & 3:** After imposing the interface & branch limits, no major change in corridor flows and generation dispatch as no constraints were binding. **Some re-dispatches seen in case – 3 due to local loadings (Ektuni ICT)**
- **Case - 4:** Under transmission outage and binding interface constraints, major change in generation dispatch, corridor flows and LMPs

5. OPF Results

NR Import Scenario

Region	Case - 1		Case - 2		Case - 3		Case - 4	
	Marginal Generator	LMP of Marginal Generator Node	Marginal Generator	LMP of Marginal Generator Node	Marginal Generator	LMP of Marginal Generator Node	Marginal Generator	LMP of Marginal Generator Node
NR	-	-	-	-	-	-	Kawai (4.95 Rs/KWh)	4.95 Rs/KWh
Local Constraint*	-	-	-	-	Koradi - II (2.17 Rs/KWh)	2.19 Rs/KWh	Koradi - II (2.17 Rs/KWh)	2.17 Rs/KWh
All India	Kahalgaon (3.70 Rs/KWh)	3.71 Rs/KWh	Kahalgaon (3.70 Rs/KWh)	3.71 Rs/KWh	Kahalgaon (3.70 Rs/KWh)	3.71 Rs/KWh	WPCL (2.82 Rs/KWh)	2.83 Rs/KWh

Local constraint: N-1 non-compliance of 765/400 kV, 2x1500 MVA Ektuni ICTs

Region	Average LMP (Rs/KWh)			
	Case - 1	Case - 2	Case - 3	Case - 4
NR	3.92	3.92	3.93	5.18
WR	3.83	3.83	3.95	3.53
SR	3.81	3.81	3.82	3.50
ER	3.77	3.77	3.74	3.55
NER	3.81	3.81	3.80	3.67

LMPs on all India Map:

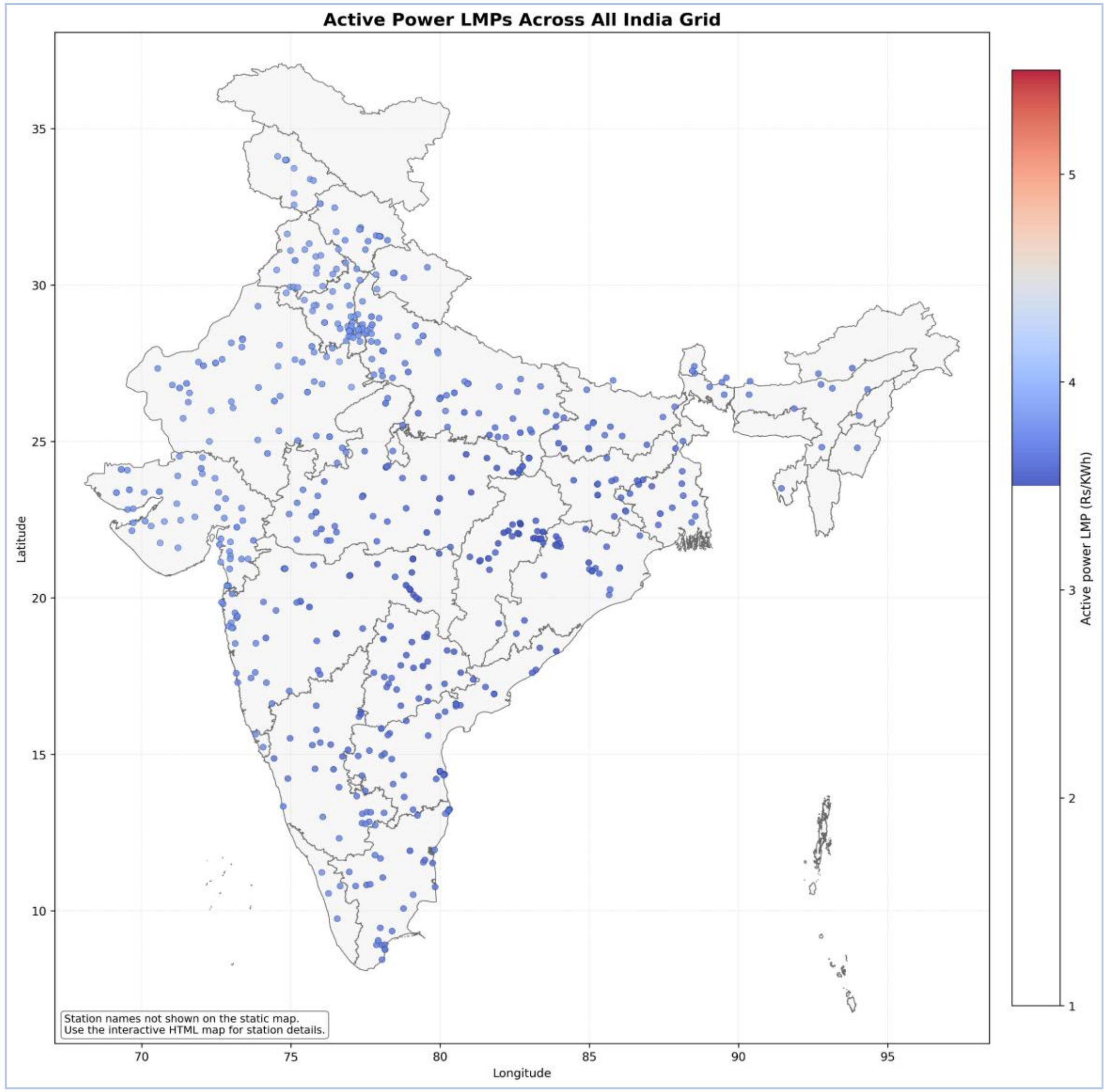
Static Maps

Interactive Maps

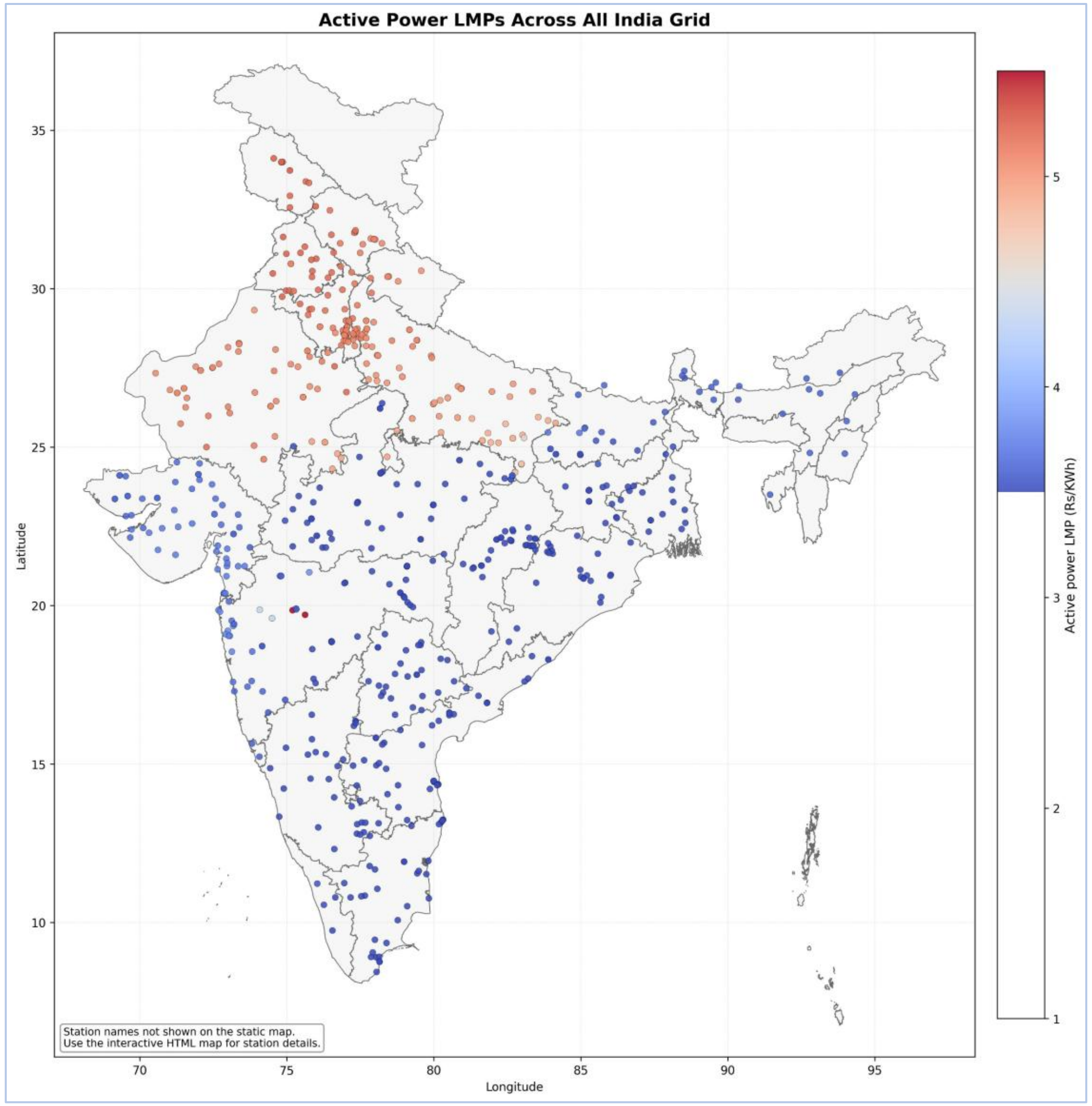
- **Base case:** NR import violations were there under the base case
- **Case – 1:** With copper plate model (no transmission constraint imposed), the NR import flow got within the TTC limits
- **Case – 2 & 3:** After imposing the interface & branch limits, no major change in corridor flows and generation dispatch as no constraints were binding. Some re-dispatches seen in case – 3 due to local loadings (Ektuni ICT)
- **Case - 4:** Under transmission outage and binding interface constraints, major change in generation dispatch, corridor flows and LMPs

5. OPF Results

NR Import Scenario



Case – 1: Copper plate model



Case – 4: Both interface & branch limit constraints imposed & binding

5. OPF Results

NR Import Scenario

<table>
<tr><th>Study Cases</th><th>Description</th><th>Link to interactive Maps</th></tr>
<tr><td>Case – 1</td><td>Voltage limits imposed + No network constraints</td><td rowspan="4">Click here to go to the link

Scenario to be chosen from the dropdown</td></tr>
<tr><td>Case – 2</td><td>Voltage limits + Interface limits imposed</td></tr>
<tr><td>Case – 3</td><td>Voltage limits + Branch flow limits imposed</td></tr>
<tr><td>Case – 4</td><td>Voltage limits + Branch flow limits + interface limits imposed

(HVDC Champa – Kurukshetra Under Outage, ~6000 MW TTC Reduction)</td></tr>
</table>

5. OPF Results

NR Import Scenario

Case-1 | NR Import Summary OPF										
Top 10 Cheapest Generators (Sorted by Variable Cost - Ascending)										
Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
324028	JAYPEE NIGRI	T1	700	385	59	650	700	50	0	315
324028	JAYPEE NIGRI	T2	700	385	59	650	700	50	0	315
352034	ACBIL	T1	30	17	109	30	30	0	0	13
354028	ACBIL	T1	135	74	109	135	135	0	0	61
354028	ACBIL	T2	135	74	109	135	135	0	0	61
254021	JITPL	T1	600	330	117	370	600	230	0	270
254021	JITPL	T2	600	330	117	370	600	230	0	270
257002	DARLIPALLI	T1	800	440	122	494	800	306	0	360
257002	DARLIPALLI	T2	800	440	122	494	800	306	0	360
354032	JPL STAGE-II	T1	600	330	133	554	600	46	0	270
			700	385	59	650	700	50	0	315
Top 10 Costliest Generators (Sorted by Variable Cost - Descending)										
Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
332199	ABHIJEET220	T1	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T2	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T3	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T4	62	34	1,083	45	34	-11	28	0
312027	UKAI-T	T3	200	110	1,080	186	110	-76	90	0
312027	UKAI-T	T5	210	115	1,080	195	116	-80	94	0
434021	UDUPI PCL	T1	600	330	1,080	545	330	-215	270	0
434021	UDUPI PCL	T2	600	330	1,080	518	330	-188	270	0
312055	APL MUNDRA	T1	330	182	1,070	237	182	-56	148	0
312055	APL MUNDRA	T2	330	182	1,070	234	182	-53	148	0

Gas, Nuclear & Off-bar thermal units not included in this table

5. OPF Results

NR Import Scenario

Case-2 | NR Import Summary OPF										
Top 10 Cheapest Generators (Sorted by Variable Cost - Ascending)										
Bus Number	**Plant name / Bus name (PSSE)**	**Unit ID**	**Maximum Capacity (Pmax)**	**Technical Minimum (0.55 of Pmax)**	**Variable Cost (P/KWh)**	**Dispatch before OPF**	**Dispatch after OPF**	**Change in Dispatch**	**Head room after OPF (Pmax - Dispatch after OPF)**	**Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)**
324028	JAYPEE NIGRI	T1	700	385	59	650	700	50	0	315
324028	JAYPEE NIGRI	T2	700	385	59	650	700	50	0	315
352034	ACBIL	T1	30	17	109	30	30	0	0	13
354028	ACBIL	T1	135	74	109	135	135	0	0	61
354028	ACBIL	T2	135	74	109	135	135	0	0	61
254021	JITPL	T1	600	330	117	370	600	230	0	270
254021	JITPL	T2	600	330	117	370	600	230	0	270
257002	DARLIPALLI	T1	800	440	122	494	800	306	0	360
257002	DARLIPALLI	T2	800	440	122	494	800	306	0	360
354032	JPL STAGE-II	T1	600	330	133	554	600	46	0	270
Top 10 Costliest Generators (Sorted by Variable Cost - Descending)										
Bus Number	**Plant name / Bus name (PSSE)**	**Unit ID**	**Maximum Capacity (Pmax)**	**Technical Minimum (0.55 of Pmax)**	**Variable Cost (P/KWh)**	**Dispatch before OPF**	**Dispatch after OPF**	**Change in Dispatch**	**Head room after OPF (Pmax - Dispatch after OPF)**	**Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)**
332199	ABHIJEET220	T1	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T2	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T3	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T4	62	34	1,083	45	34	-11	28	0
312027	UKAI-T	T3	200	110	1,080	186	110	-76	90	0
312027	UKAI-T	T5	210	115	1,080	195	116	-80	94	0
434021	UDUPI PCL	T1	600	330	1,080	545	330	-215	270	0
434021	UDUPI PCL	T2	600	330	1,080	518	330	-188	270	0
312055	APL MUNDRA	T1	330	182	1,070	237	182	-56	148	0
312055	APL MUNDRA	T2	330	182	1,070	234	182	-53	148	0

Gas, Nuclear & Off-bar thermal units not included in this table

5. OPF Results

NR Import Scenario

Case-3 | NR Import Summary OPF										
Top 10 Cheapest Generators (Sorted by Variable Cost - Ascending)										
Bus Number	**Plant name / Bus name (PSSE)**	**Unit ID**	**Maximum Capacity (Pmax)**	**Technical Minimum (0.55 of Pmax)**	**Variable Cost (P/KWh)**	**Dispatch before OPF**	**Dispatch after OPF**	**Change in Dispatch**	**Head room after OPF (Pmax - Dispatch after OPF)**	**Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)**
324028	JAYPEE NIGRI	T1	700	385	59	650	700	50	0	315
324028	JAYPEE NIGRI	T2	700	385	59	650	700	50	0	315
352034	ACBIL	T1	30	17	109	30	30	0	0	13
354028	ACBIL	T1	135	74	109	135	135	0	0	61
354028	ACBIL	T2	135	74	109	135	135	0	0	61
254021	JITPL	T1	600	330	117	370	600	230	0	270
254021	JITPL	T2	600	330	117	370	600	230	0	270
257002	DARLIPALLI	T1	800	440	122	494	800	306	0	360
257002	DARLIPALLI	T2	800	440	122	494	800	306	0	360
354032	JPL STAGE-II	T1	600	330	133	554	600	46	0	270
Top 10 Costliest Generators (Sorted by Variable Cost - Descending)										
Bus Number	**Plant name / Bus name (PSSE)**	**Unit ID**	**Maximum Capacity (Pmax)**	**Technical Minimum (0.55 of Pmax)**	**Variable Cost (P/KWh)**	**Dispatch before OPF**	**Dispatch after OPF**	**Change in Dispatch**	**Head room after OPF (Pmax - Dispatch after OPF)**	**Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)**
332199	ABHIJEET220	T1	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T2	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T3	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T4	62	34	1,083	45	34	-11	28	0
312027	UKAI-T	T3	200	110	1,080	186	110	-76	90	0
312027	UKAI-T	T5	210	115	1,080	195	116	-80	94	0
434021	UDUPI PCL	T1	600	330	1,080	545	330	-215	270	0
434021	UDUPI PCL	T2	600	330	1,080	518	330	-188	270	0
312055	APL MUNDRA	T1	330	182	1,070	237	182	-56	148	0
312055	APL MUNDRA	T2	330	182	1,070	234	182	-53	148	0

Gas, Nuclear & Off-bar thermal units not included in this table

5. OPF Results

NR Import Scenario

Case-4 | NR Import Summary OPF										
Top 10 Cheapest Generators (Sorted by Variable Cost - Ascending)										
Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
324028	JAYPEE NIGRI	T1	700	385	59	650	700	50	0	315
324028	JAYPEE NIGRI	T2	700	385	59	650	700	50	0	315
352034	ACBIL	T1	30	17	109	30	30	0	0	13
354028	ACBIL	T1	135	74	109	135	135	0	0	61
354028	ACBIL	T2	135	74	109	135	135	0	0	61
254021	JITPL	T1	600	330	117	370	600	230	0	270
254021	JITPL	T2	600	330	117	370	600	230	0	270
257002	DARLIPALLI	T1	800	440	122	494	800	306	0	360
257002	DARLIPALLI	T2	800	440	122	494	800	306	0	360
354032	JPL STAGE-II	T1	600	330	133	554	600	46	0	270
Top 10 Costliest Generators (Sorted by Variable Cost - Descending)										
Bus Number	Plant name / Bus name (PSSE)	Unit ID	Maximum Capacity (Pmax)	Technical Minimum (0.55 of Pmax)	Variable Cost (P/KWh)	Dispatch before OPF	Dispatch after OPF	Change in Dispatch	Head room after OPF (Pmax - Dispatch after OPF)	Leg room after OPF (Dispatch after OPF - 0.55 of Pmax)
332199	ABHIJEET220	T1	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T2	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T3	62	34	1,083	45	34	-11	28	0
332199	ABHIJEET220	T4	62	34	1,083	45	34	-11	28	0
312027	UKAI-T	T3	200	110	1,080	186	110	-76	90	0
312027	UKAI-T	T5	210	115	1,080	195	116	-80	94	0
434021	UDUPI PCL	T1	600	330	1,080	545	330	-215	270	0
434021	UDUPI PCL	T2	600	330	1,080	518	330	-188	270	0
312055	APL MUNDRA	T1	330	182	1,070	237	182	-56	148	0
312055	APL MUNDRA	T2	330	182	1,070	234	182	-53	148	0

Gas, Nuclear & Off-bar thermal units not included in this table

Ongoing and Future Work

1. Preparation of monthly all-India PSS/E base and OPF cases from RA results (in addition to manual cases) in coordination with RLDCs

2. Integration of all ~650 units & other data (demand, RE, Hydro etc.) with SCADA keys (currently mapped only with RA output) for preparation of accurate all India real-time cases

3. Anticipation of corridor flows for regions and states in advance

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